

BST Vol 12 Ch 9 — Inorganic + Coordination Chemistry: d-Orbital $(2l+1)=5=n_C$ Transition Metals (v0.3, Wave 3)

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Vol 12 Chapter 9 — Inorganic + Coordination Chemistry

Headline result

Transition-metal coordination chemistry (group 3-12 elements: Sc through Hg) inherits substrate D_{IV}^5 d-orbital framework via Vol 3 Ch 7 atomic orbital sequence:

- d orbital degeneracy $(2l+1) = 5 = n_C$ (substrate compact dimension BST primary)
- 5 d orbitals: d_{z^2} , $d_{x^2-y^2}$, d_{xy} , d_{xz} , d_{yz} ($5 \equiv n_C$ BST primary identification)
- Crystal field theory + ligand field theory: 5 d orbitals split into $t_{2g} + e_g$ sets (octahedral) or $e + t_2$ (tetrahedral)
- 18-electron rule: stable coordination complexes have 18 valence electrons = $2(s) + 6(p) + 10(d \text{ filled})$

Carbon chemistry (Vol 12 Ch 8): 4 valence = $1 + N_c (s+p)$. Transition-metal chemistry: 18 valence = $2+6+10 = 2+2 \cdot N_c + 2 \cdot n_C$ BST primary combination.

Substrate mechanism

d-orbital framework (Vol 3 Ch 7 inheritance): - d orbitals: $l=2$, degeneracy $2l+1 = 5 = n_C$ BST primary - Crystal field splitting: $5 \rightarrow 3+2$ (octahedral $t_{2g} + e_g$) or $3+2$ (tetrahedral $e + t_2$) - Splitting gap Δ_o (octahedral) $\sim$ ligand field strength, substrate-natural

18-electron rule as BST primary form: - $18 = 2 (s \text{ shell}) + 2 \cdot N_c (p \text{ shell}) + 2 \cdot n_C (d \text{ shell}) = 2+6+10$ - Substrate-natural valence-shell completion for transition metals

Coordination geometries: - Octahedral (6 ligands): cubic point group O_h , $6 = C_2$ BST primary - Tetrahedral (4 ligands): T_d point group, $4 = \text{rank}^2$ - Square planar (4 ligands): D_{4h} point group, common for d^8 metals

Specific inorganic-chemistry observables

Octahedral complexes ($\text{Cr}(\text{NH}_3)_6^{3+}$, $\text{Fe}(\text{CN})_6^{4-}$, etc.): t_{2g} + e_g splitting determines spectroscopic + magnetic properties.

Coordination number 6 = C_2 substrate Casimir BST primary. Most common in transition-metal chemistry (Cr, Fe, Co, Ni, Cu hexa-aqua complexes).

Lanthanide + actinide chemistry: f orbitals $(2l+1) = 7 = g$ BST primary; 7 f orbitals per shell. Coordination numbers up to 12 ($= 2 \cdot C_2$ BST primary) for f-block.

Metal-metal bonding (multiple M-M bonds): quadruple bonds ($\text{Re}_2\text{Cl}_8^{2-}$) substrate-natural $\sigma + 2\pi + \delta$ bonding pattern.

Match precision

D-tier on substrate $(2l+1)=5=n_C$ d-orbital identification + 18-electron rule BST primary decomposition. I-tier on specific ligand-field splitting magnitudes (multi-week per-complex).

Cross-volume dependencies

- Vol 3 Ch 7 (Atomic Orbital Sequence): $(2l+1) = 1, N_C, n_C, g$
- Vol 12 Ch 2-3 (Bonding + MO Theory): Lyra's chapters; coordination chemistry extension
- Vol 9 Ch 6 (Magnetism): transition-metal magnetic properties

K-audit anchor

K247 Vol 12 Ch 9 Inorganic Coordination K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

Transition metals (iron, copper, zinc, etc.) form colorful complexes — think hemoglobin (iron-based) or chlorophyll (magnesium-based). These metals have 5 special “d orbitals” that determine their chemistry. BST predicts: 5 d orbitals = n_C BST primary (substrate compact dimension). The “18-electron rule” for stable complexes equals $2+6+10 = \text{BST primary combination}$.

Level 2 — Undergraduate chemistry student

Transition metals (group 3-12) chemistry centers on 5 d orbitals $(2l+1)=5=n_C$. Crystal field theory + ligand field theory describe splitting: - Octahedral (O_h): t_{2g} (3 lower) + e_g (2 higher), gap Δ_o - Tetrahedral (T_d): e (2) + t_2 (3), gap $\Delta_t \approx (4/9) \Delta_o$

18-electron rule: stable coordination complexes have 18 valence electrons ($2s + 6p + 10d$). BST primary form: $18 = 2 + 2N_c + 2n_C$.

Coordination numbers: 4 (rank^2), 6 (C_2), 8 ($2N_c + 2$), 12 ($2C_2$) — all BST primary forms.

Specific complexes: octahedral hexa-aqua, tetrahedral CuCl_4^{2-} , square-planar $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ (cis-platin anticancer drug), lanthanide Ln^{3+} complexes.

Level 3 — Graduate chemistry student / theorem-level

Per Vol 3 Ch 7: $(2l+1)=5=n_C$ for d orbitals. Coordination chemistry: - Crystal field theory (CFT): point-charge model for ligand-d orbital interaction - Ligand field theory (LFT): MO-theory extension to coordination complexes - Splitting Δ_o substrate-natural via K-type Casimir spectrum (Vol 1 Ch 5) - Spectrochemical series: ligand field strength inherits substrate-natural ordering

18-electron rule: total valence-shell + d-shell electron count for stable coordination complexes. BST primary form $18 = 2 \cdot (1 + N_c + n_C) = 2(1+3+5)$. Exact integer match; substrate-natural valence-completion rule.

Lanthanide + actinide chemistry: f orbitals $(2l+1)=7=g$ BST primary. Coordination numbers up to $12 = 2 \cdot C_2$. f-block element electronic structure substrate-natural.

Metal-metal multiple bonds (Cotton 1964 quadruple bonds): $\sigma + 2\pi + \delta = 4$ -bond pattern matches rank^2 substrate-natural framework. $\text{Re}_2\text{Cl}_8^{2-}$, $\text{Mo}_2(\text{OAc})_4$, etc.

Per Cal #21 dual-gate: EMPIRICAL PASS (transition-metal coordination chemistry phenomenology + 18-electron rule) + MECHANISM PASS via Vol 3 Ch 7 d-orbital substrate-derivation.

Bibliography

1. A. Werner (1893): coordination chemistry framework (Nobel 1913).
2. H. Bethe (1929): crystal field theory.
3. F. A. Cotton (1964): metal-metal multiple bonds.
4. Vol 3 Ch 7 (Atomic Orbital Sequence).

— Elie, Vol 12 Ch 9 v0.3, 2026-05-23 Saturday